# `moria_dev`

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**Abstract**

This note documents major changes to some of the scripts used in `01.XYM` and `03.LOC_TRANS` on the `moria_dev` branch of MORIA. The `moria_dev` branch is designed to work with two-exposure datasets ( 2×F814W + 2×F606W). It also allows for cases with a single missing F606W exposure (2×F814W + 1×F606W).

# Contents

# 1 Overview

```
00.DATA/{F814W,F606W}/*_flc.fits
        |
        v  hst1pass  (reduce_wfc3.src  OR  reduce_wfc3_1exp.src)
        |
01.XYM/*.F814W_xympquvwrd , *.F606W_xympquvwrd
        |
        +-- 2+2 --> dex_no_gaia.e
        +-- 1+2 --> 1exp_no_gaia.e
        |
        v
        |
MATCHUP / stacks / dex outputs
        |
        v  LOC_TRANS
        |
        +-- 2+2 --> xym2mat_new.e
        +-- 1+2 --> 1exp_xym2mat_new.e
        |
        v
        |
04.EXTRACT_PSF --> 06.FIT --> 07.CALIBRATION
```

## 1.1 2+2 datasets

moria_dev assumes inputs include two F814W _flc.fits exposures and two F606W _flc.fits exposures. Catalogs are named *.F814W_xympquvwrdand *.F606W_xympquvwrd. The script used to generate the output stack is dex_no_gaia.e and xym2mat_new.e.

## 1.2 1+2 datasets

moria_dev assumes inputs include two F814W _flc.fits exposures and one F606W _flc.fits exposures. Catalogs are named *.F814W_xympquvwrdand *.F606W_xympquvwrd. The script used to generate the output stack is 1exp_no_gaia.e (fork of dex_no_gaia) and 1exp_xym2mat_new.e (fork of xym2mat_new)

# 2 Fortran Programs

After hst1pass finishes, each exposure has its own star list in that exposure's detector coordinates.

## 2.1 hst1pass.F

| Script | Mode | What it does |
|---|---|---|
| `reduce_wfc3.src` | 2+2 | Run `hst1pass` on every F606W and F814W `_flc`, rename outputs to `*.F606W_xympquvwrd` / `*.F814W_xympquvwrd` |
| `reduce_wfc3_1exp.src` | 1+2 | Same commands; used when only one F606W flc exists (glob naturally finds one V file) |
| `reduce_acs.src` | 2+2 | ACS version (ACS PSF + official GDC names) |
| `reduce_acs_1exp.src` | 1+2 | ACS single-F606W path |

## 2.2 dex_no_gaia.F to dex_no_gaia.e

### 2.2.1 Input

The inputs include four files output after running hst1pass, i.e., the four `*.xympquvwrd`files. `dex_no_gaia.F` only accepts four files. The program scans arguments for names that contain both `F606W/F814W` and `*.xympquvwrd`. It stores:

- `NARG_V1`, `NARG_V2` — the two F606W lists,
- `NARG_I1`, `NARG_I2` — the two F814W lists.

The four lists are ordered as

- N=1 F606W-1
- N=2 F606W-2
- N=3 F814W-1
- N=4 F814W-2

### 2.2.2 How the code works

For each of the four `*.xympquvwrd`catalogs, `dex_no_gaia.F` reads star positions, magnitudes, filters, exposure times, epochs and chip boundaries. It writes this information into `dex_no_gaia_STEP02_A.nim_info`.

Then, from the RA/Dec footprints of the exposures, it computes a sky box and its center (RMIN/RMAX, DMIN/DMAX and RCEN, DCEN). The center becomes the focal point of the final master frame.

Instead of reading a Gaia file, `dex_no_gaia.F` sets NREF =1 (the first F606W exposure). It selects bright stars from that list only.

```
do L = 1, Ls_n(NREF)
   if (m_ln(L,NREF).lt.-11) then   ! bright instrumental mag
      Gs = Gs + 1
      ra_g(Gs) = r_ln(L,NREF)      ! use that star's RA/Dec
      de_g(Gs) = d_ln(L,NREF)
```

```
      ... fill Gaia-like arrays with dummy PM/parallax/magnitude
   endif
enddo
```

The arrays originally reserved for the Gaia catalog are filled with HST stars that look like Gaia entries. This is a mock internal reference list. Proper motions are set to zero and the magnitudes used are placeholders. This approach works because we treat the first 606W exposure as a temporary sky catalog. Bright, well-measured stars on that exposure define a smaller but accurate local reference. Every other exposure is then tied to that same set of stars. You don't really need Gaia positions for stacking and relative photometry.

The reference stars are mapped to the UV gaia grid centered on (RCEN and DCEN). within 0.050"/pixel. `dex_no_gaia_STEP05_A.uvm_gaia` is written at this point.

Then, each exposure is matched to the internal reference list. For every exposure $N = 1 \dots 4$, we find which of its bright stars ($m < -11$) coincide with the internal reference and linearly transform each exposure to the coordinates of the internal reference list. Using $m < -11$ keeps the reference clean (bright, high S/N) and matches the cut used later when matching exposures.

#### 2.2.3 Creating the F606W and F814W stacks

We merge detections across exposures into one master star list, then refine it (combine F606W and F814W photometry where both exist).

For each filter (`V1I2 = 1` for F606W, `2` for F814W) the code:

1. Locates the matching `_flc.fits` under `../00.DATA/F606W/` or `F814W/`.
2. Reads the ACS/WFC3 chips into a $4096 \times 4096$ buffer (`readfits_r4e` for extensions 1 and 4).
3. Scales each exposure by `EXPT/340` so different exposure times are comparable.
4. Uses the instrument distortion file (`STDGDC_WFC3UV_*.fits` or ACS official GDC) plus the linear transformations from matching each exposure to the internal reference list to map every pixel $(u, v)$ in the internal lsit back to detector $(x_r, y_r)$.
5. Samples the flc at that detector pixel (nearest integer pixel; skips the chip gap near $y \approx 2048.5$).
6. Combines the two exposures per filter: if both contribute, take the average of the two scaled values; if only one contributes, use that value.
7. Write `dex_no_gaia_STEP10A_F606W.fits` and texttdex_no_gaia_STEP10B_F814W.fits.

### 2.3 1exp_no_gaia.F to 1exp_no_gaia.e

Most of this code is similar to `dex_no_gaia.F`. We still build an internal reference from exposure `NREF = 1` with $m < -11$. The internal reference only needs one bright exposure. The only difference is that reserves only one `NARG` slot for F606W. The three lists are ordered as:

- N=1 F606W-1
- N=2 F814W-1
- N=3 F814W-2

### 2.4 xym2mat_new.F and 1exp_xym2mat_new.F

These are similar to `dex_no_gaia.F` and `1exp_no_gaia.F`, but instead of running in `01.XYM`, they run in `03.LOC_TRANS`. Instead of using the first F606W exposure to build an internal reference star list, they use `NEARBY_REF_STARS.XYIVB_targ`.